

LECTURE 9: MORE ON CLT: OTHER PROOFS AND COMPLEMENTARY RESULTS

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

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NON-WEAK LIMITS

Why can we only hope for a weak limit in the CLT setting? Let $X_1, X_2, \dots$ be i.i.d. random variables with mean zero and variance one.

Claim:

$$\lim_{n \rightarrow \infty} \frac{X_1 + X_2 + \dots + X_n}{\sqrt{n}} \in \sigma(X_{N+1}, X_{N+2}, \dots) \text{ for any } N \in \mathbb{Z}_+.$$

so it belongs to the tail σ -algebra.

Why?

$$\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_N}{\sqrt{n}} + \frac{X_{N+1} + \dots}{\sqrt{n}} \stackrel{\text{a.s.}}{=} \lim_{n \rightarrow \infty} \frac{X_{N+1} + \dots}{\sqrt{n}}.$$

By the Kolmogorov 0–1 law, any almost sure limit of this sequence must be a constant (if it exists), and it is not (one can observe, for example, that the variance does not converge to zero). One can also show that the limsup of this sequence (which always exists) is ∞ (see below). If the sequence converges in probability, we can select an a.s. converging subsequence to obtain a contradiction.

To show that the limsup of the sequence diverges, let $S_n = X_1 + \dots + X_n$. By the Kolmogorov 0–1 law, $\limsup_n S_n/\sqrt{n} = \infty$ occurs with probability either zero or one (note that the limsup always exists). By the CLT, we have $\lim_n \mathbb{P}(S_n/\sqrt{n} > x) > 0$ for all $x \in \mathbb{R}$. By the reverse Fatou's lemma (noting that $\mathbb{1}_{\{S_n/\sqrt{n} > x\}}$ is dominated by the constant one function), we have $\mathbb{P}(\limsup_n S_n/\sqrt{n} > x) \geq \limsup_n \mathbb{P}(S_n/\sqrt{n} > x) > 0$. This implies that $\limsup_n S_n/\sqrt{n} > x$ almost surely for all $x \in \mathbb{R}$, and therefore we conclude that $\limsup_n S_n/\sqrt{n} = \infty$ a.s.

LINDEBERG'S PROOF OF THE CLT (LINDEBERG EXCHANGE METHOD)

Let's discuss one more proof of the CLT that gives an explanation as to why there is a universal limit in the CLT.

Theorem 9.1 (CLT again). *Let $X_1, X_2, \dots$ be i.i.d. random variables with mean zero, variance one, and finite third moment $E|X_i|^3 \leq C$ (this can be relaxed). Then*

$$\sum_{i=1}^n \frac{X_i}{\sqrt{n}} \xrightarrow{w} N(0, 1).$$

Proof (Lindeberg exchange method). Our goal is to show that for all continuous and bounded functions g ,

$$\mathbb{E} \left[g \left(\sum_{i=1}^n \frac{X_i}{\sqrt{n}} \right) \right] \rightarrow \mathbb{E} \left[g \left(\sum_{i=1}^n \frac{Z_i}{\sqrt{n}} \right) \right] \quad \text{as } n \rightarrow \infty,$$

where $Z_i \sim N(0, 1)$ are i.i.d. standard normal random variables.

Swapping trick: let

$$\begin{aligned} Y_n^0 &:= X_1 + X_2 + X_3 + \dots + X_n, \\ Y_n^1 &:= Z_1 + X_2 + X_3 + \dots + X_n, \\ Y_n^3 &:= Z_1 + Z_2 + X_3 + \dots + X_n, \\ &\vdots \\ Y_n^n &:= Z_1 + Z_2 + Z_3 + \dots + Z_n. \end{aligned}$$

Observe that we can write

$$g \left(\frac{Y_n^0}{\sqrt{n}} \right) - g \left(\frac{Y_n^n}{\sqrt{n}} \right) = \sum_{j=1}^n \left\{ g \left(\frac{Y_n^{j-1}}{\sqrt{n}} \right) - g \left(\frac{Y_n^j}{\sqrt{n}} \right) \right\}$$

as a telescoping sum. Furthermore, for each j , we have

$$Y_n^{j-1} = R_j + X_j \quad \text{and} \quad Y_n^j = R_j + Z_j,$$

where $R_j = Z_1 + \dots + Z_{j-1} + X_{j+1} + \dots + X_n$, and R_j, X_j, Z_j are mutually independent.

By using a third order Taylor expansion for g that is sufficiently differentiable (a subtlety: we may approximate any continuous, bounded g by smooth functions), we obtain

$$\mathbb{E}[g(R + X)] = \mathbb{E}[g(R)] + \mathbb{E}[X \cdot g'(R)] + \mathbb{E} \left[\frac{X^2}{2!} g''(R) \right] + \mathbb{E} \left[\frac{X^3}{3!} g'''(R') \right]$$

for some $R' \in (R, R + X)$ (mean-value form of the remainder). By applying this to $R = \frac{R_j}{\sqrt{n}}$ and $X = \frac{X_j}{\sqrt{n}}$ and using independence of R and X , we obtain

$$\begin{aligned} \mathbb{E} \left[g \left(\frac{R_j + X_j}{\sqrt{n}} \right) \right] &= \mathbb{E} \left[g \left(\frac{R_j}{\sqrt{n}} \right) \right] + \frac{1}{\sqrt{n}} \underbrace{\mathbb{E}[X_j]}_{=0} \mathbb{E} \left[g' \left(\frac{R_j}{\sqrt{n}} \right) \right] + \frac{1}{2n} \underbrace{\mathbb{E}[X_j^2]}_{=1} \mathbb{E} \left[g'' \left(\frac{R_j}{\sqrt{n}} \right) \right] \\ &\quad + \frac{1}{6n^{3/2}} \underbrace{\mathbb{E}[X_j^3]}_{\leq C} \mathbb{E} \left[\underbrace{g''' \left(\frac{R'_j}{\sqrt{n}} \right)}_{\leq a} \right]. \end{aligned}$$

Similarly, we may apply the same calculation with $X = \frac{Z_j}{\sqrt{n}}$, using the fact that $\mathbb{E}[Z_j] = 0$, $\mathbb{E}[Z_j^2] = 1$, and $\mathbb{E}[Z_j^3] = 0$ to obtain an analogous expression (where the first three terms are

identical and will cancel out). Therefore, to conclude,

$$\begin{aligned} \left| \mathbb{E} \left[g \left(\frac{Y_n^0}{\sqrt{n}} \right) - g \left(\frac{Y_n^n}{\sqrt{n}} \right) \right] \right| &\leq \sum_{j=1}^n \mathbb{E} \left| g \left(\frac{Y_n^{j-1}}{\sqrt{n}} \right) - g \left(\frac{Y_n^j}{\sqrt{n}} \right) \right| \\ &\leq \sum_{j=1}^n \frac{C}{n^{3/2}} = \frac{C}{\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned} \quad \square$$

Let's note here that different proofs of the CLT have different advantages, and each extend to further results in probability:

- *Characteristics functions* proof ("Fourier-analytic method")
- *Lindeberg exchange* proof (with swapping trick)
- *Moment method* ($\lim_{n \rightarrow \infty} \mathbb{E}[X_n^k] = \mathbb{E}[X^k]$ for all k – beyond the scope of this class.)
- ...

There are also more direct proofs for special cases, like the *de Moivre–Laplace CLT* for the Binomial distribution; this involves a direct computation of the distribution of

$$\sum_{i=1}^n \frac{X_i - \frac{1}{2}}{\sqrt{n/2}},$$

where X_i are 0.5-Bernoulli random variables, as n grows.

CLT FOR NON-I.I.D. RANDOM VARIABLES

Can we relax the assumptions to show a CLT for random variables that are not i.i.d.?
Consider a *triangular array* of random variables:

$$\begin{array}{llll} \text{independent} & \rightarrow & X_{11} & \\ \text{independent} & \rightarrow & X_{21} & X_{22} \\ \text{independent} & \rightarrow & X_{31} & X_{32} & X_{33} \\ \text{independent} & \rightarrow & X_{41} & X_{42} & X_{43} & X_{44} \\ \dots & & \dots & & \dots & \end{array}$$

Theorem 9.2 (Lindeberg CLT). *For each n , let $\{X_{n,k}\}_{1 \leq k \leq n}$ be a collection of independent random variables with finite variance. Let $\mu_{n,k} := \mathbb{E}[X_{n,k}]$, $\sigma_{n,k}^2 := \text{Var}(X_{n,k})$, and $s_n^2 := \sum_{k=1}^n \sigma_{n,k}^2$. Suppose that the random variables satisfies Lindeberg's condition: for any $\varepsilon > 0$,*

$$L_n(\varepsilon) := \sum_{k=1}^n \mathbb{E}[\tilde{X}_{n,k}^2 \cdot \mathbf{1}_{|\tilde{X}_{n,k}| > \varepsilon}] \xrightarrow{n \rightarrow \infty} 0, \quad \text{where } \tilde{X}_{n,k} := \frac{X_{n,k} - \mu_{n,k}}{s_n}.$$

Then

$$\sum_{k=1}^n \frac{X_{n,k} - \mu_{n,k}}{s_n} \xrightarrow{w} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

We won't prove this theorem in class. It can be proved via the characteristic functions argument; see, e.g., [Chatterjee, Section 8.11].

Some observations:

- Since $L_n(\varepsilon) \rightarrow 0$ as $n \rightarrow \infty$ by the Lindeberg condition for any $\varepsilon > 0$,

$$\text{Var}(\tilde{X}_{n,k}) \leq \mathbb{E}[\tilde{X}_{n,k}^2] \leq \mathbb{E}[\tilde{X}_{n,k}^2 \cdot \mathbf{1}_{|\tilde{X}_{n,k}| > \varepsilon}] + \varepsilon^2 \leq L_n(\varepsilon) + \varepsilon^2 \sim_{n \rightarrow \infty} \varepsilon^2.$$

So,

$$\max_{1 \leq k \leq n} \text{Var}(\tilde{X}_{n,k}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Intuition: *individual contributions of summands are small.*

- Theorem 9.2 generalizes the CLT for i.i.d. random variables. Indeed, if $\{X_i\}_{i \geq 1}$ is a sequence of i.i.d. random variables with $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i^2 = 1$, then we can consider the triangular array $X_{n,k} = \frac{X_k}{\sqrt{n}}$, which satisfies the Lindeberg condition:

$$L_n(\varepsilon) = n \cdot \mathbb{E} \left[\frac{X_1^2}{n} \mathbf{1}_{\{|X_1| > \varepsilon \sqrt{n}\}} \right] \xrightarrow{\text{DCT}} 0 \quad \text{as } n \rightarrow \infty.$$

- The Lindeberg condition is sufficient but, in general, not necessary: for example, for each n , consider the (independent) random variables $X_{n,1} \sim N(0, \frac{1}{2})$ and $X_{n,k} \sim N(0, \sigma_{n,k}^2)$ for $k \geq 2$ such that $\sum_{k=2}^n \sigma_{n,k}^2 = \frac{1}{2}$. Then the contribution of the first random variable is not uniformly small, but the CLT holds.

NON-GAUSSIAN LIMITS

What about other limiting distributions? The $N(0, 1)$ limit is the result of proper scaling of the components. With careful scaling, other limiting distributions are possible, including the following:

Theorem 9.3 (Poisson limit). *For each n , let $\{X_{n,k}\}_{1 \leq k \leq n}$ be a collection of independent, non-negative integer-valued random variables. That is, $X_{n,k} \in \mathbb{Z}_{\geq 0}$, and denote $p_{n,k} := \mathbb{P}(X_{n,k} = 1)$ and $\varepsilon_{n,k} := \mathbb{P}(X_{n,k} \geq 2)$. Suppose that*

$$\sum_{k=1}^n p_{n,k} \rightarrow \lambda < \infty, \quad \max_{1 \leq k \leq n} p_{n,k} \rightarrow 0, \quad \text{and} \quad \sum_{k=1}^n \varepsilon_{n,k} \rightarrow 0$$

as $n \rightarrow \infty$. Then

$$Z_n := \sum_{k=1}^n X_{n,k} \xrightarrow{\text{w}} \text{Poi}(\lambda).$$

Interpretation: the random variable Z_n “counts the occurrences of rare events”: the number of events grows but the probability that each of them is non-zero tends to zero.

Proof sketch. Step 1 (truncation): Let $\tilde{Z}_n := \sum_{k=1}^n X_{n,k} \mathbf{1}_{\{X_{n,k} \leq 1\}}$. Note that

$$\mathbb{P}(\tilde{Z}_n \neq Z_n) \leq \sum_{k=1}^n \mathbb{P}(X_{n,k} > 1) = \sum_{k=1}^n \varepsilon_{n,k} \rightarrow 0$$

by assumption. Hence, $\tilde{Z}_n - Z_n \xrightarrow{p} 0$, so it is enough to show that $\tilde{Z}_n \Rightarrow \text{Poi}(\lambda)$. (Think through the last claim)

Step 2 (characteristic functions): By independence,

$$\varphi_{\tilde{Z}_n}(t) = \prod_{k=1}^n \varphi_{X_{n,k}}(t) = \prod_{k=1}^n (1 - p_{n,k} + p_{n,k} e^{it}).$$

The characteristic function of a $\text{Poisson}(\lambda)$ random variable is given by $\varphi_{\text{Poi}(\lambda)}(t) = e^{\lambda(e^{it}-1)}$ (exercise). Since $\lambda = \lim_{n \rightarrow \infty} \sum_{k=1}^n p_{n,k}$ by assumption, it is enough to show that

$$L := \prod_{k=1}^n (1 - p_{n,k}(1 - e^{it})) - \prod_{k=1}^n e^{p_{n,k}(e^{it}-1)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Claim (exercise): for all $p \in (0, 1)$,

$$|e^p(e^{it} - 1)| \leq 1 \quad \text{and} \quad |1 + p(e^{it} - 1)| \leq 1.$$

Recall the following result from the proof of the CLT from the previous lecture: if $z_1, z_2, \dots, z_n \in \mathbb{C}$ and $w_1, w_2, \dots, w_n \in \mathbb{C}$ are such that $|z_i|, |w_i| \leq \theta$ for every i , then

$$\left| \prod_{j=1}^n z_j - \prod_{j=1}^n w_j \right| \leq \theta^{n-1} \sum_{j=1}^n |z_j - w_j|.$$

Therefore, applying this result, we obtain

$$\begin{aligned} L &\leq \sum_{k=1}^n |1 + p_{n,k}(e^{it} - 1) - e^{p_{n,k}(e^{it}-1)}| \\ &\leq \sum_{k=1}^n p_{n,k}^2 |e^{it} - 1|^2 \quad (\text{exercise: power series for complex exponential}) \\ &\leq 4 \sum_{k=1}^n p_{n,k}^2 \leq 4(\max_k p_{n,k}) \sum_{k=1}^n p_{n,k} \rightarrow 0. \end{aligned} \quad \square$$

RATE OF CONVERGENCE FOR THE CLT

So far, we only know that $P(Z_n \leq t) \rightarrow P(Z \leq t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx$ as $n \rightarrow \infty$, where $Z \sim N(0, 1)$ and $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ (X_i mean zero, variance one).

But we do not know how quick is the convergence. In particular, *how can we estimate the error for large but finite n ?* (That is, in the *non-asymptotic regime*?) This is nontrivial.

Theorem 9.4 (Berry-Esseen). *Let $X_1, X_2, \dots$ be i.i.d. random variables with mean zero, variance one, and finite third moment: $\rho := \mathbb{E}|X_1|^3 < \infty$. Let $Z_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$. Then for any $t \in \mathbb{R}$,*

$$|\mathbb{P}(Z_n \leq t) - \mathbb{P}(Z \leq t)| \leq \frac{C\rho}{\sqrt{n}},$$

where $Z \sim N(0, 1)$ and C is an absolute constant.

Remarks:

- We won't prove this – it can be proved using Fourier-analytic methods (e.g., see Theorem 2.2.8 of Terence Tao's textbook, *Topics in Random Matrix Theory*), as well as a probabilistic technique known as Stein's method (e.g., see Section 11.1 of these lecture notes by Tomasz Tkocz, or Section 3 of the textbook *Normal Approximation by Stein's Method* by Chen, Goldstein, and Shao).

- *Sharpness of the convergence rate:* consider $X_n \sim \pm 1$ symmetric random variables, and $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$. By symmetry,

$$\tilde{p} = P(Z_n \leq 0) = P(Z_n \geq 0) = \frac{1}{2}.$$

Furthermore, $2\tilde{p} - P(Z_n = 0) = 1$. So,

$$\tilde{p} = \frac{1 + P(Z_n = 0)}{2} = \frac{1}{2} + \frac{1}{2} \binom{n}{n/2} 2^{-n}.$$

Thus, noting that $P(Z \leq 0) = 1/2$ for $Z \sim N(0, 1)$,

$$|P(Z_n \leq 0) - P(Z \leq 0)| = \frac{1}{2} \binom{n}{n/2} 2^{-n} \sim \frac{1}{2} \frac{1}{\sqrt{\pi n}}.$$

is of the right order $O(1/\sqrt{n})$ as guaranteed by Theorem 9.4. Here, we used Stirling's approximation $n! = (1 + O(n^{-1})) \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$ to approximate the binomial coefficient by $\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n}} \left(\frac{n}{e}\right)^n$. (Note that $\frac{(2k)!}{k!k!} \sim \frac{1}{\sqrt{\pi k}} \left(\frac{2k}{e}\right)^{2k} \left(\frac{e}{k}\right)^{2k} = \frac{2^{2k}}{\sqrt{\pi k}}.$)

- *Another example on the convergence rate:* consider i.i.d. $X_n \sim \text{Bernoulli}(p)$ random variables. Then (after centering and scaling),

$$\begin{aligned} \rho &= \mathbb{E} \left| \frac{X_1 - p}{\sqrt{p(1-p)}} \right|^3 = \frac{p(1-p)^3 + (1-p)p^3}{(p(1-p))^{3/2}} \\ &= \frac{p(1-p)(1-2p+p^2+p^2)}{(p(1-p))^{3/2}} = \frac{1-2p(1-p)}{\sqrt{p(1-p)}}. \end{aligned}$$

- If $np \sim \delta$ is constant (i.e., n is large, $p \sim \delta/n$ is small), then

$$\frac{C\rho}{\sqrt{n}} \sim \frac{C}{\sqrt{np(1-p)}} \sim \frac{C}{\delta} = O(1)$$

is of constant order; thus, Theorem 9.4 does not give a good probability bound. *Reason:* Gaussian approximation is not tight here – in this regime, the Poisson limit (Theorem 9.3) is the right scaling!

- Compare this with $p = \frac{1}{2}$:

$$\sum_{i=1}^n \frac{(X_i - \frac{1}{2})}{\frac{1}{2}\sqrt{n}} = \frac{(\sum_{i=1}^n X_i) - \frac{1}{2}n}{\sqrt{n \cdot \frac{1}{2} \cdot \frac{1}{2}}} = \frac{S_n - pn}{\sqrt{np(1-p)}} \xrightarrow{w} N(0, 1).$$

This is the *de Moivre–Laplace CLT* for the Binomial distribution, which can be proved directly using Stirling's approximation.